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Studierichting: Informatica
Oefeningenreeks 5D nr. 4.22

$$y_1 = 2 - \sqrt{x} \text{ en } y_2 = \frac{\sqrt{x} + 1}{2\sqrt{x}}$$

1. Snijpunten

$$2 - \sqrt{x} = \frac{\sqrt{x} + 1}{2\sqrt{x}}$$

$$2\sqrt{x}(2 - \sqrt{x}) = \sqrt{x} + 1$$

$$4\sqrt{x} - 2x = \sqrt{x} + 1$$

$$3\sqrt{x} - 2x - 1 = 0$$

$$t = \sqrt{x}$$

$$3t - 2t^2 - 1 = 0 \iff -2t^2 + 3t - 1 = 0$$

$$t = \frac{-3 \pm \sqrt{9 - 4 \cdot (-2) \cdot (-1)}}{-4}$$

$$t = \frac{-3 \pm \sqrt{1}}{-4}$$

$$t = 1 \text{ of } t = \frac{1}{2}$$

$$\sqrt{x} = 1 \Rightarrow x = 1$$

$$\sqrt{x} = \frac{1}{2} \Rightarrow x = \frac{1}{4}$$

2. Bepaal bovenste functie

$$\text{Neem } x = \frac{1}{2}$$

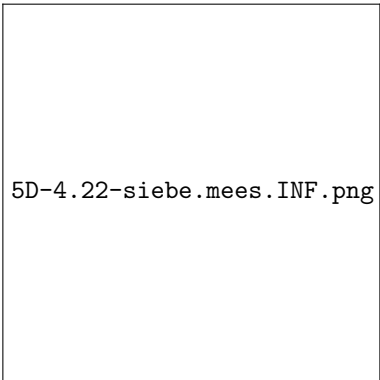
$$y_1 = 2 - \sqrt{\frac{1}{2}} \approx 1,29$$

$$y_2 = \frac{\sqrt{\frac{1}{2}} + 1}{2\sqrt{\frac{1}{2}}} \approx 1,21$$

$$\Rightarrow y_1 \text{ boven } y_2$$

⇒ Tekening

x	0	
y_1	2	+
y_1'	-	
y_1''	+	
y_1	$\searrow$	
	)	
x	0	
y_2	$+\infty$	-
y_2'	-	
y_2''	+	
y_2	$\searrow$	
	)	



References